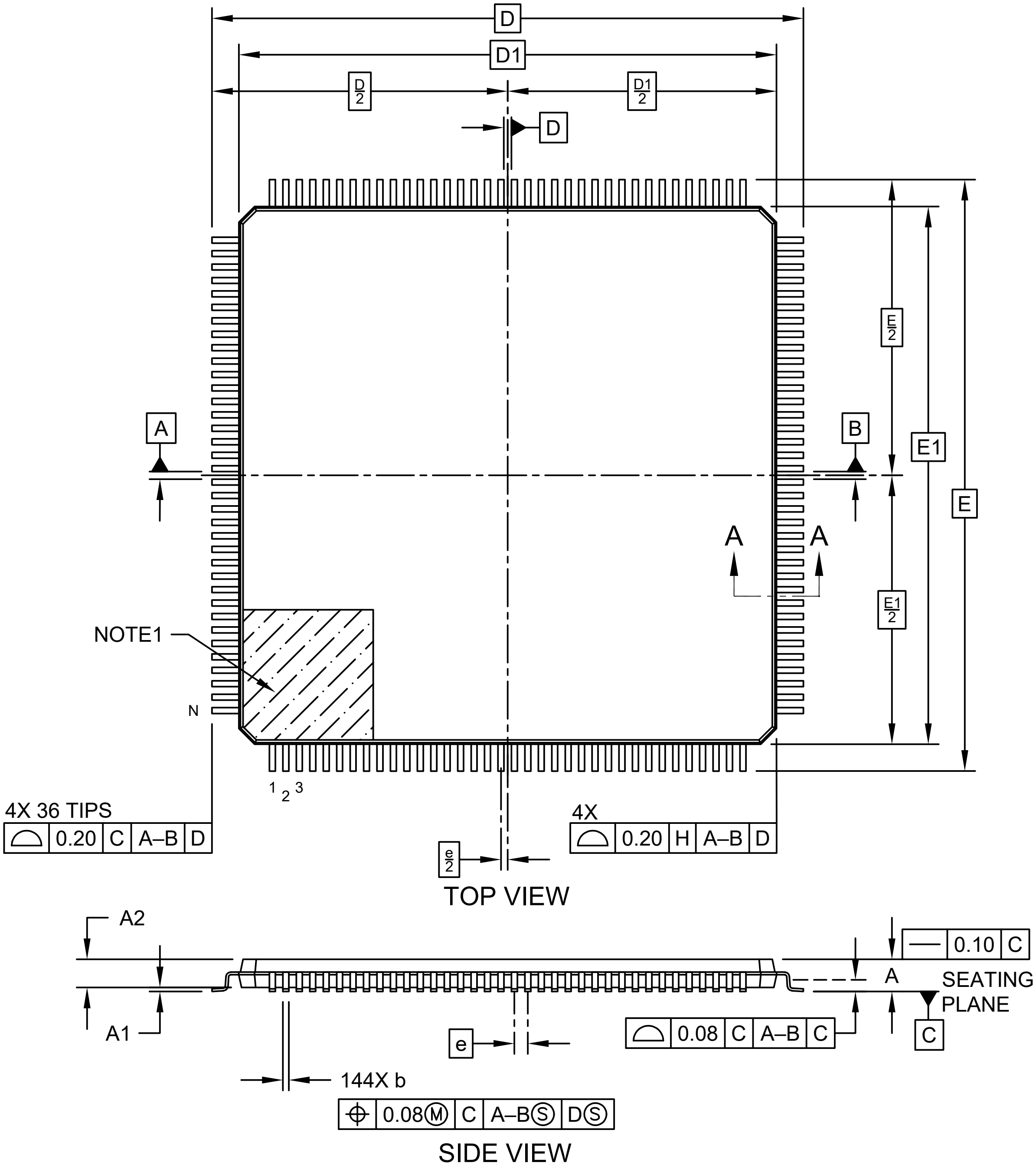
144-Lead Plastic Thin Quad Flat Pack (Z8X) - 20x20x1.0 mm Body [TQFP] SMSC Legacy Package VTQE3

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



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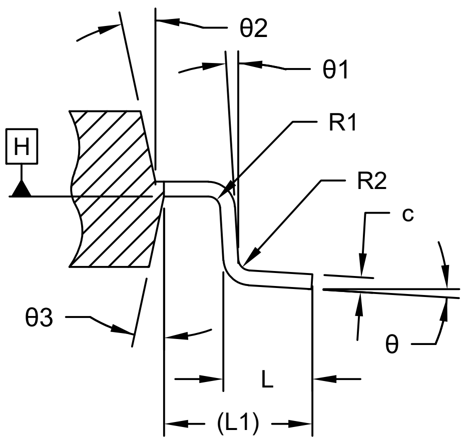
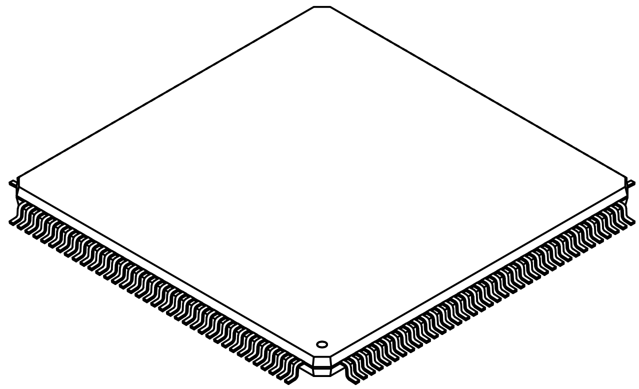
144-Lead Plastic Thin Quad Flat Pack (Z8X) - 20x20x1.0 mm Body [TQFP]
SMSC Legacy Package VTQE3

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



SECTION A–A



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		144		
Pitch	e		0.50 BSC		
Overall Height	A		-	-	1.20
Standoff	A1		0.05	-	0.15
Molded Package Thickness	A2		0.95	1.00	1.05
Overall Length	D		22.00 BSC		
Molded Package Length	D1		20.00 BSC		
Overall Width	E		22.00 BSC		
Molded Package Width	E1		20.00 BSC		
Terminal Width	b		0.17	0.22	0.27
Terminal Thickness	c		0.09	-	0.20
Terminal Length	L		0.45	0.60	0.75
Footprint	L1		1.00 REF		
Lead Bend Radius	R1		0.08	-	-
Lead Bend Radius	R2		0.08	-	0.20
Foot Angle	θ		0°	3.5°	7°
Lead Angle	θ1		0°	-	-
Mold Draft Angle	θ2		11°	12°	13°
Mold Draft Angle	θ3		11°	12°	13°

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

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144-Lead Plastic Thin Quad Flat Pack (Z8X) - 20x20x1.0 mm Body [TQFP]
SMSC Legacy Package VTQE3

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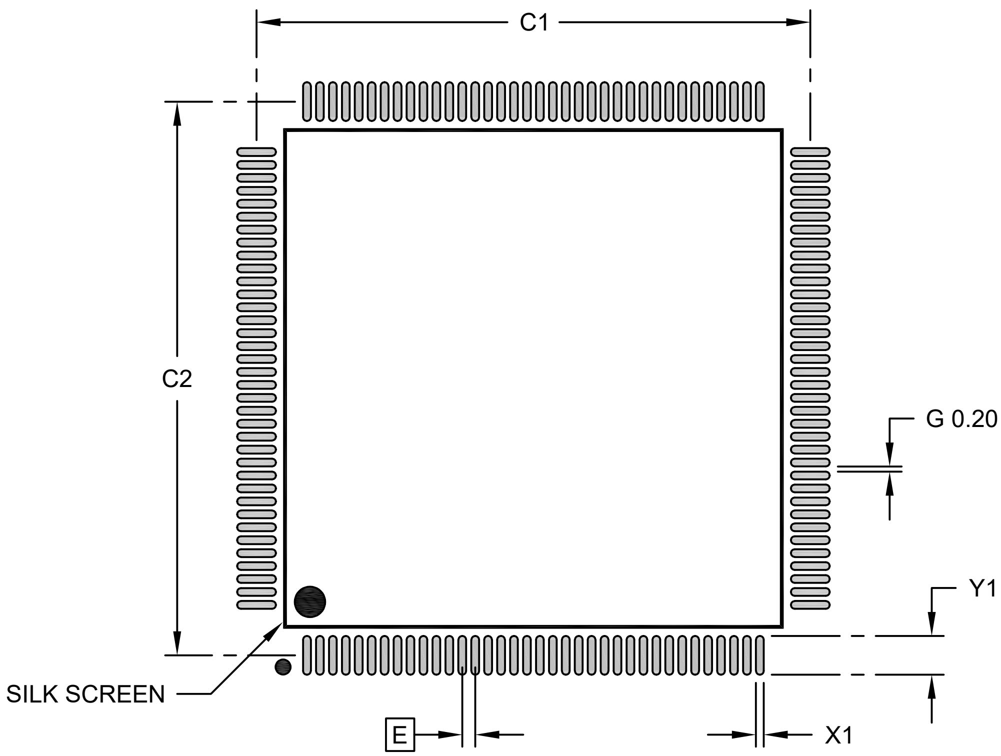

RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Contact Pad Spacing	C1		21.40	
Contact Pad Spacing	C2		21.40	
Contact Pad Width (Xnn)	X1			0.30
Contact Pad Length (Xnn)	Y1			1.50
Contact Pad to Contact Pad (Xnn)	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

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